Revenue Recovery Strategy for Big Mountain Resort

Sook hyun Lee

Data Science Guided Capstone Project, Oct 2025

Springboard

Problem

Context

- New chairlift added to improve visitor distribution
- Raises seasonal operating costs by **$1.54M**

Objective

- Fully recover added costs within **2 years**

Revenue Strategies

1. **Premium Pricing**
   - Slightly above market average
2. **Operational Reallocation**
   - Shift resources to high-margin services, No ticket price change

Recommendation and Key Findings

Immediate Action:

- Increase ticket price by **at least $1 (less than $10)**

Future Strategy:

- **Temporarily close a couple of runs** for one year
- Monitor **visitor numbers** and **net revenue**
- Gradually close additional runs **until net revenue growth plateaus**

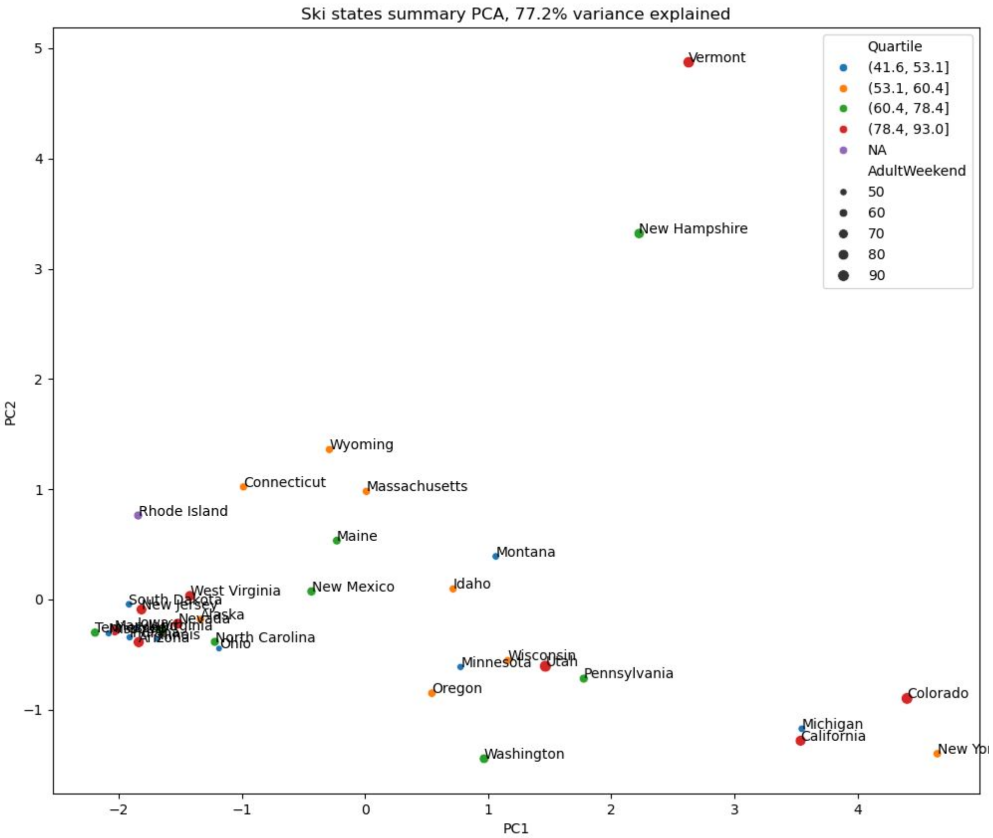
Goal:

- Align pricing with market rates
- Increase revenue **without added costs**

Data

- **Ski data**
  - 277 ski resort records and 25 fields after cleaning
  - Name, location, facilities, ticket price and operational information
- **State data**
  - State population and area statistics

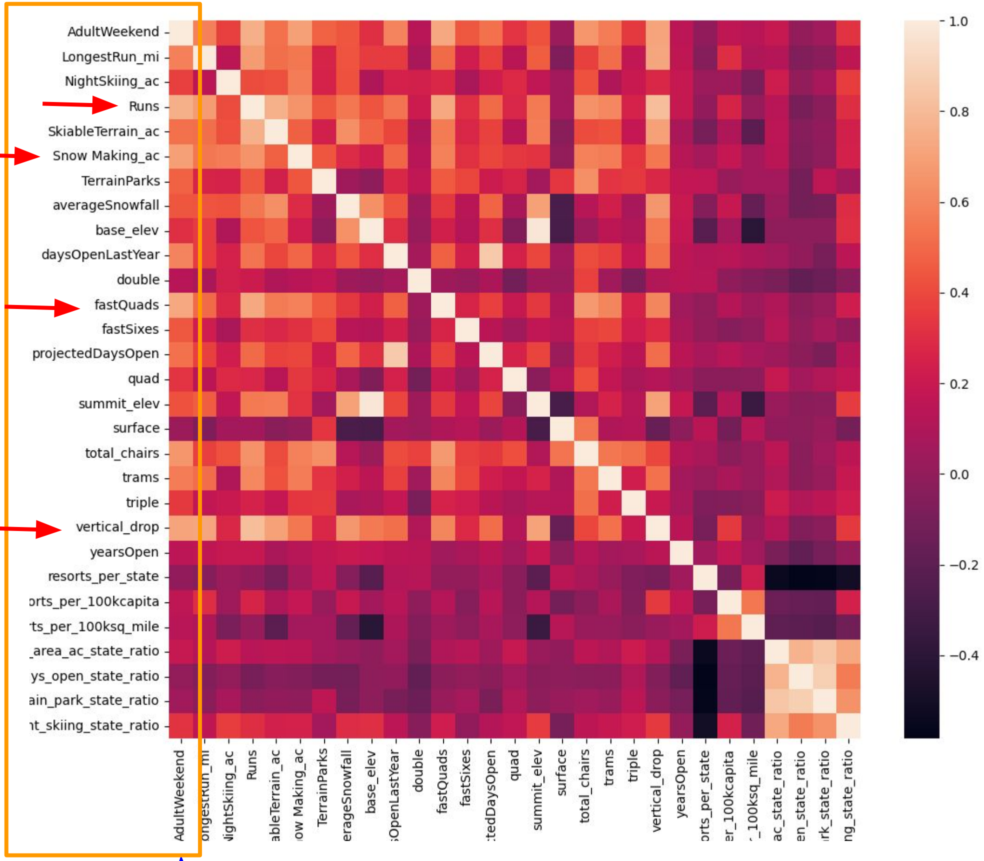
- **Target variable:**
  - AdultWeekend



PCA analysis and **average ticket prices by state** show **no** compelling reason to treat any **state differently**.

Which Feature Are Highly Correlated with Target Variable?

- Runs
- Snow Making ac
- Fast Quads
- Vertical drop



Target variable: **AdultWeekend**

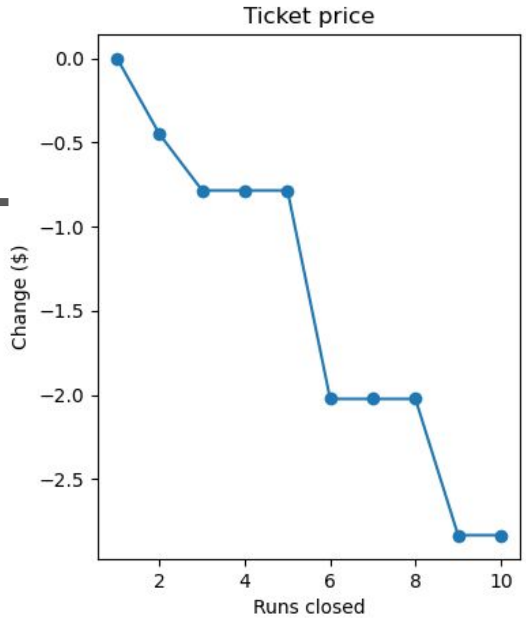
Modeling and Results

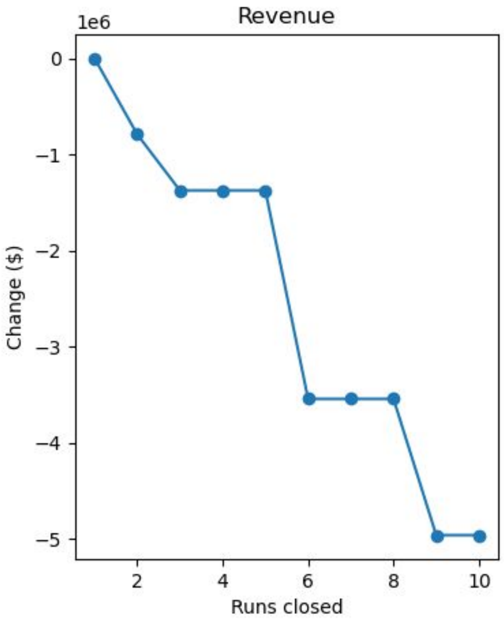
- **Evaluation metric:**
  Mean Absolute Error (MAE)
- **Models:**
  - **Baseline**: Average ticket price in training dataset, MAE (test) = **20**
  - **Linear Regression**: MAE (test) = **11.8**
  - **Random Forest**: MAE (test) = **9.7**

Modeling Scenarios

Closing runs **:**

Effectively close gap between '**current**' and **'predicted**' ticket prices





Scenarios:

A. Close up to 10 of the least used runs.

B. Adding a run, increasing the vertical drop by 150 feet, and installing an additional chair lift. → Increase support for ticket price by $0.76

C. B + add 2 acres of snow making. → No change

D. Increase the longest run by .2 miles and add 4 acres of snow making capability. → No change

Summary and Future Work

Analysis Summary

- Explored how feature adjustments impact **ticket pricing**
    - **Changing number of runs** is most effective approach for recovering revenue.
- All changes must consider effects on:
    - **Visitor numbers and Operational costs**

Data Gaps for Deeper Insight

- Cost-related factors:
    - **Transportation**, **lodging**, **dining**, and **amenities**
- Business insights needed:
    - **Profit margins**
    - **Current pricing strategy**

Next Step

- Investigate why model predicts **higher optimal price** than current rates